BASE CHANGE OF A STANDARD PH COVERING IS A STANDARD PH COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2533, 2539] and proof(s) [[2541, 2551]]; this spec covers the complete environment instead (statement 2533-2539, proof span(s) [(2541, 2551)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original definition 0DBD: sources/stacks-chow/tags.txt, source lines 16693–16693. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01S1: sources/stacks-chow/tags.txt, source lines 2322–2322. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W4: sources/stacks-chow/tags.txt, source lines 2469–2469. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\{f_j : U_j \to T\}_{j=1,\ldots,m}$ be a standard ph covering. Let $T' \to T$ be a morphism of affine schemes. Then $\{U_j \times_T T' \to T'\}_{j=1,\ldots,m}$ is a standard ph covering.*

Proof. Let $f : U \to T$ be proper surjective and let an affine open covering $U = \bigcup_{j=1,\ldots,m} U_j$ be given as in Definition 0DBD. Then $U \times_T T' \to T'$ is proper surjective (Morphisms, Lemmas 01S1 and 01W4). Also, $U \times_T T' = \bigcup_{j=1,\ldots,m} U_j \times_T T'$ is an affine open covering. This concludes the proof. □

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